

FORM TP 2010233



TEST CODE **02138020**

MAY/JUNE 2010

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PHYSICS

UNIT 1 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **SIX** questions.
2. Section A consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the spaces provided in this question paper.
3. Section B consists of **THREE** questions. Candidates must attempt **ALL** questions in this section. Answers for this section must be written in the space at the end of each question.
4. All working **MUST** be **CLEARLY** shown.
5. The use of non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.

LIST OF PHYSICAL CONSTANTS

Universal gravitational constant	G	=	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Acceleration due to gravity	g	=	9.8 m s^{-2}
1 Atmosphere	Atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Boltzmann's constant	k	=	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Density of water		=	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water		=	$4200 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific latent heat of fusion of ice		=	$3.34 \times 10^5 \text{ J kg}^{-1}$
Specific latent heat of vaporization of water		=	$2.26 \times 10^6 \text{ J kg}^{-1}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$
Molar gas constant	R	=	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Stefan-Boltzmann's constant	s	=	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$

SECTION A

Answer ALL questions in this section. Write your answers in the spaces provided in this answer booklet.

1. (a) Figure 1 shows a ball placed at the top of an inclined plane with a block rigidly fixed to the bottom of the plane.

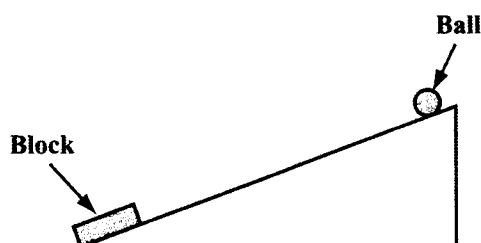


Figure 1

The ball of mass 0.60 kg is released from rest from the top of the plane and v , the velocity of the ball down the plane, is measured for various values of time t . Table 1 shows the variation of velocity with time.

Table 1

Velocity $v/\text{m s}^{-1}$	Time, t/s	Velocity $v/\text{m s}^{-1}$	Time, t/s
0	0	- 3.4	1.25
0.6	0.2	- 2.8	1.4
1.4	0.4	- 2.1	1.6
2.1	0.6	- 1.4	1.8
2.8	0.8	- 0.8	2.0
3.5	1.0	0	2.2
4.2	1.2		

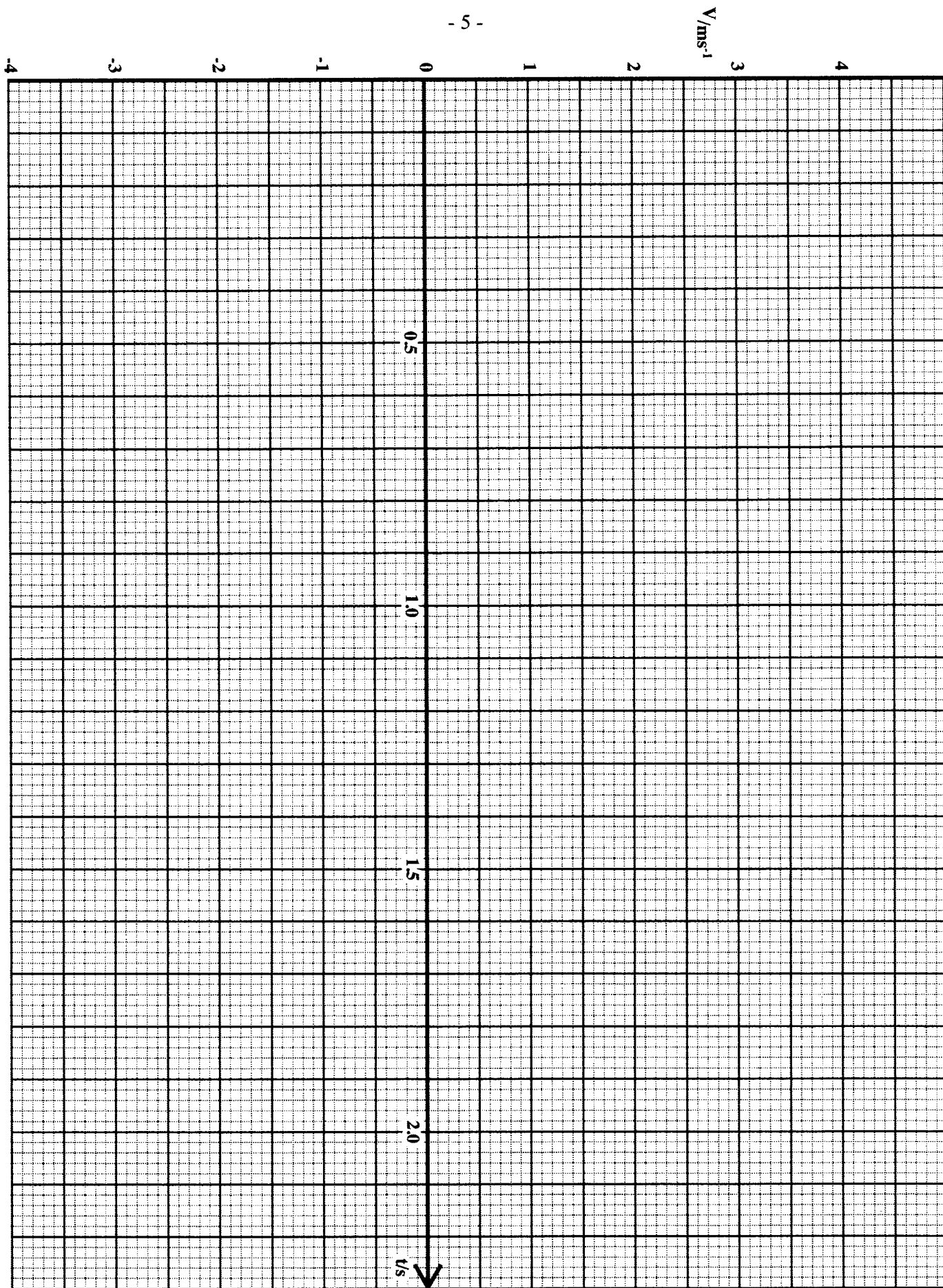
- (i) On the grid on page 5, plot a graph of velocity v , versus time t .

[4 marks]

- (ii) Using your graph, describe qualitatively the motion of the ball.

[3 marks]

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(iii) Calculate

a) the acceleration of the ball down the inclined plane

[2 marks]

b) the length of the incline

[2 marks]

c) the MEAN force experienced by the ball during the impact with the block.

[2 marks]

(iv) State, with a reason, whether the collision between the block and the ball is elastic or not.

[2 marks]

Total 15 marks

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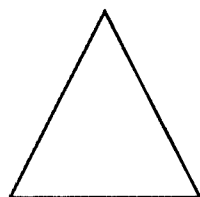
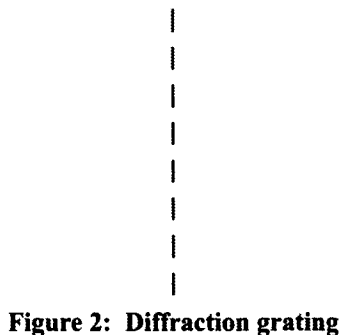
NOTHING HAS BEEN OMITTED

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2. (a) Draw rays to show the passage of **white** light through

- (i) Figure 2, a diffraction grating
- (ii) Figure 3, a triangular glass prism
- (iii) Figure 4, a rectangular glass block.

[4 marks]



(b) The graph on page 9 shows the relationship between the sines of the angle of incidence, θ_1 , and the angle of refraction, θ_2 for monochromatic yellow light travelling from glass to air.

- (i) Use the graph to find the missing values of θ_1 and θ_2 and insert them in the table below.

θ_1	$\sin \theta_1$	θ_2	$\sin \theta_2$
31.0°			
		75.2°	
		90.0°	

State the value of the critical angle of the glass. _____

[7 marks]

- (ii) Describe what happens when the angle of incidence θ_1 is 55°.

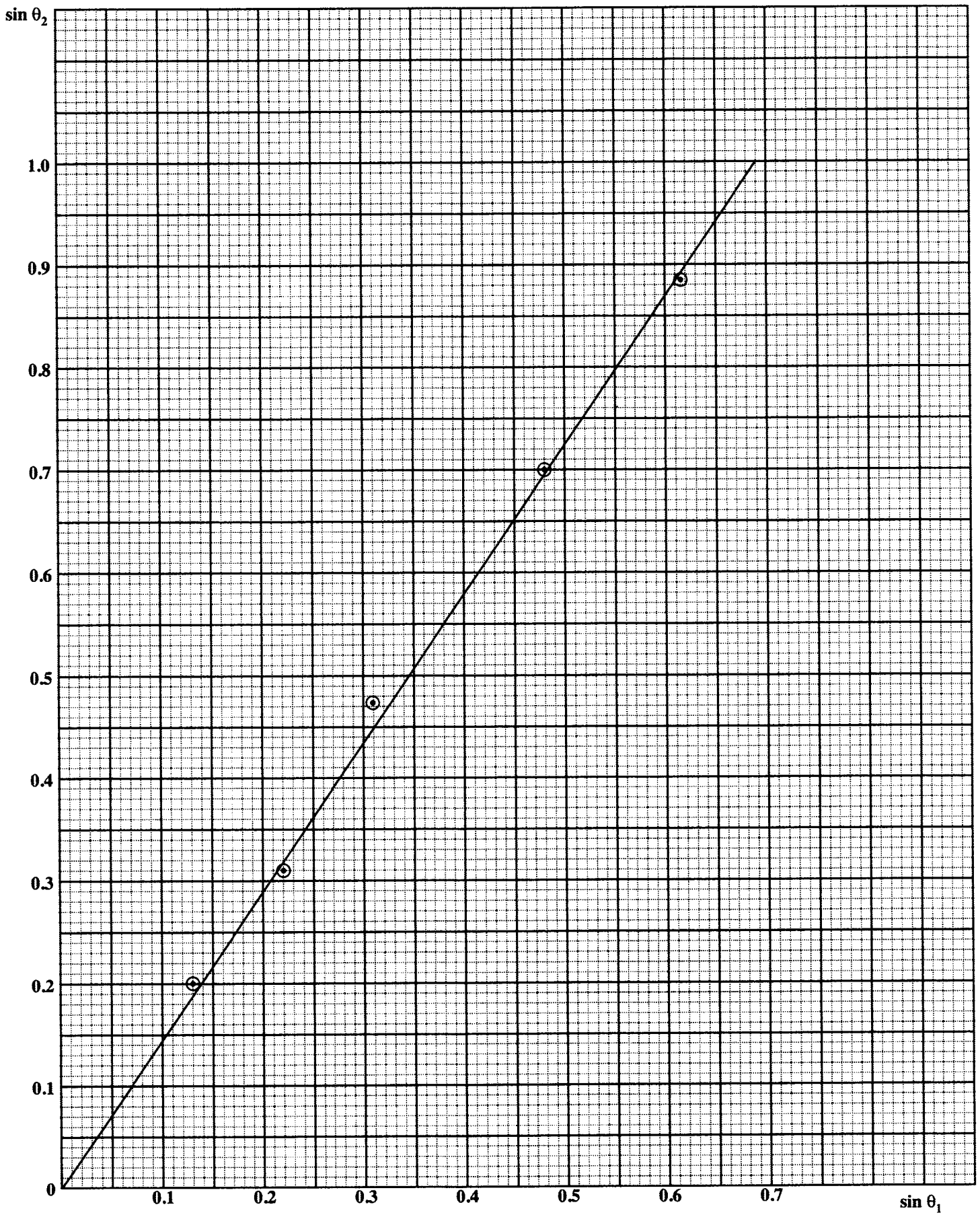
[1 mark]

- (iii) Use the gradient of the graph to determine the refractive index of the glass for this colour light.

[3 marks]

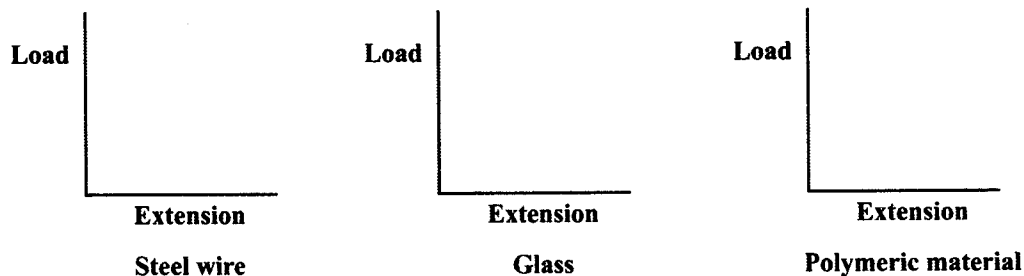
Total 15 marks

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3. (a) (i) In the spaces provided, sketch graphs of load versus extension for a steel wire, glass and a polymeric material



[3 marks]

- (ii) Define the terms 'stress' and 'strain'.

Stress:

[1 mark]

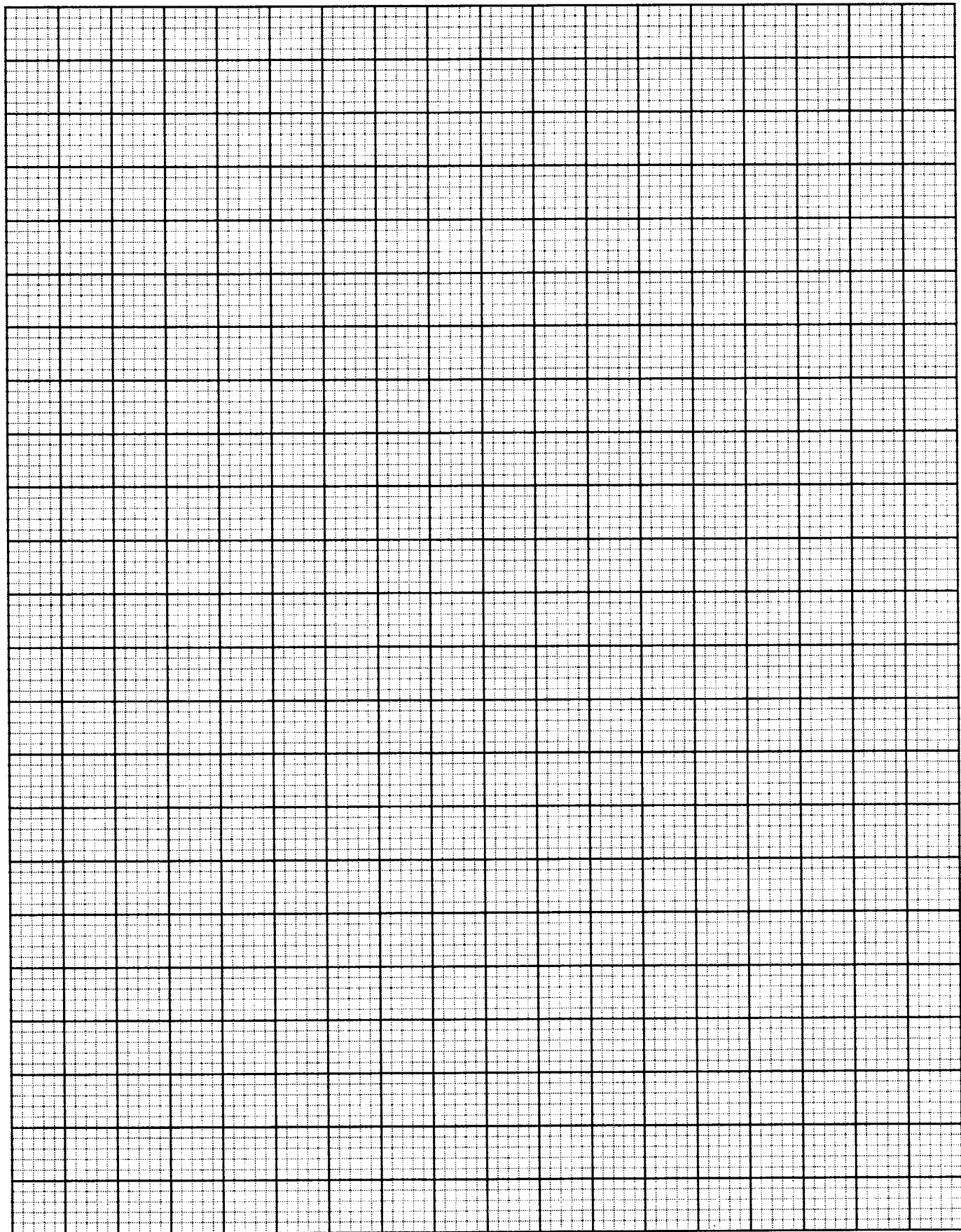
Strain:

[1 mark]

- (b) When a rubber strip with a cross-section of $3 \times 10^{-3} \text{ m}$ by $1.5 \times 10^{-3} \text{ m}$ is suspended vertically and various masses are attached to it, a student obtains the following data for length versus load.

Load, M/kg	0	0.1	0.2	0.3	0.4	0.5
Length, L/cm	5.0	5.6	6.2	6.9	7.8	10.0
Extension, $\Delta L / \text{m}$	0					

- (i) Fill in the missing values of extension, ΔL , in the table. [1 mark]
- (ii) On the grid on page 11, draw a graph of load versus extension. [4 marks]



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- (iii) Write an equation relating M and ΔL for small loads to Young's modulus E for the rubber.

Write an equation relating Young's modulus and the gradient of your graph for small loads.

[2 marks]

- (iv) Use your graph to determine Young's modulus for the rubber for small loads.

[4 marks]

Total 15 marks

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SECTION B

Answer ALL questions in this section.

Write your answers in the spaces provided at the end of each question.

4. (a) (i) State the conditions necessary for a body to be in equilibrium under the action of coplanar forces.
- (ii) A block weighing 150 N hangs from a cord. It is pulled aside, as shown in Figure 5, by a horizontal force F . Find the tension in the cord and the magnitude of F .
[7 marks]

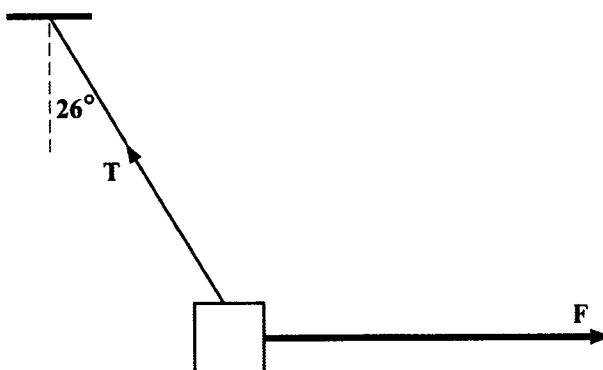


Figure 5

- (b) A boy jumps off a flat-bed truck that is 2.2 m high. The initial velocity of the boy is 1.6 m s^{-1} at an angle 20° above the horizontal.
- (i) State the horizontal and vertical components of the initial velocity.
- (ii) Calculate the time taken for the boy to reach the ground.
- (iii) How far horizontally from the truck does the boy land?

[8 marks]

Total 15 marks

Write your answer to Question 4 here.

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Write your answer to Question 4 here.

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5. Use the data presented in Figure 6 to answer this question.

(a) Figure 6 shows how the intensity and intensity level of the human ear vary with frequency.

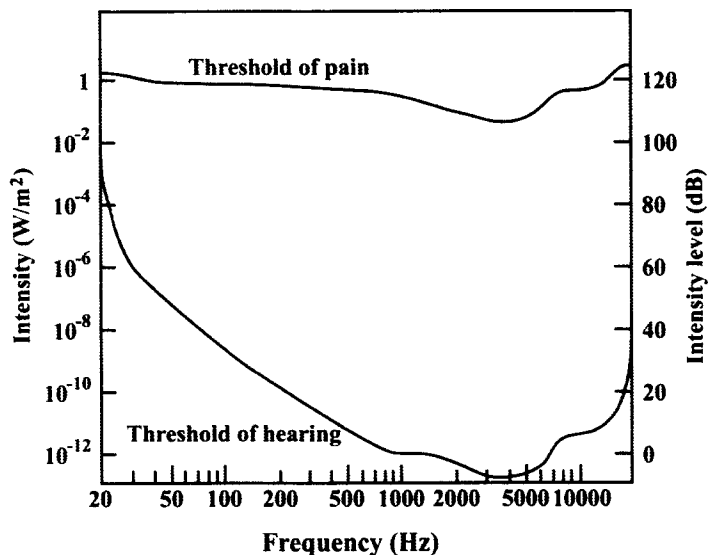


Figure 6

- (i) Explain what is meant by 'threshold of hearing' and 'threshold of pain'.
- (ii) What property of the human ear makes the decibel (dB) scale particularly useful?
- (iii) Write down an expression that relates the sound intensity I , to the intensity level β , in dB.
- (iv) What is the intensity level of a sound with intensity 3.82 mW m^{-2} ?
- (v) Figure 6 is drawn for a typical human ear. Suggest how the figure might change as a person ages.

[7 marks]

- (b) Figure 7 shows a loudspeaker connected to an audio-frequency signal generator/amplifier that is set up in front of a large flat wall. A small microphone moved between the speaker and the wall detects regions of low and high intensity.

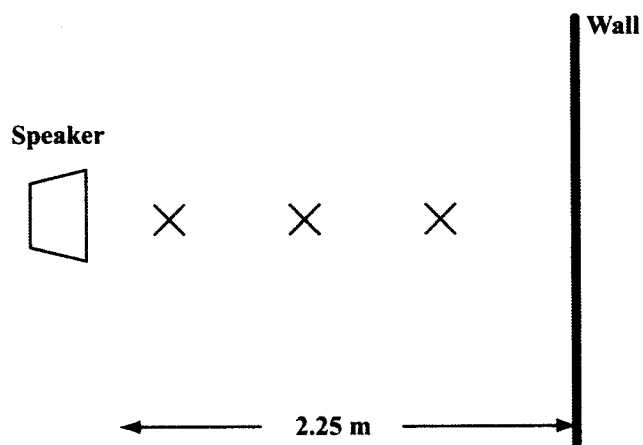


Figure 7

- (i) Explain why there are positions between the speaker and the wall where intensity is a minimum and why these minima do NOT actually have zero intensity.
- (ii) The points labelled X on Figure 7 are the only three points of minimum intensity detected at a certain frequency setting. What is the frequency?
- (iii) When the signal generator is set at 165 Hz how far from the wall is the last **maximum** intensity position?

[Velocity of sound = 330 m s^{-1}]

[8 marks]

Total 15 marks

Write your answer to Question 5 here.

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6. (a) (i) A hot object emits electromagnetic waves. Explain in terms of the kinetic theory how this radiation is able to warm a distant cold body.
- (ii) The Earth's surface is said to be warming up due to the accumulation of carbon dioxide and other gases in the atmosphere. Explain this 'greenhouse effect'.

[5 marks]

- (b) A black woodstove with a total surface area 4.6 m^2 is made from cast iron $4.0 \times 10^{-3} \text{ m}$ thick. The interior wall of the stove is at 650°C while the exterior wall is at 647°C . The temperature of the surrounding air is 30°C .

Calculate

- (i) the rate of the heat conduction through the stove wall
- (ii) the net rate of heat loss by radiation from the stove, assuming it acts as a black body
- (iii) the heat the stove loses by a combination of conduction and convection in the surrounding air. Explain your answer.

(Thermal conductivity of cast iron, $k = 80.4 \text{ Wm}^{-1}\text{K}^{-1}$)

[10 marks]

Total 15 marks

Write your answer to Question 6 here.

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